



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

COMPRESSED AIR SCRUBBER (D-2207)

Docu. No. : 14780-8120-PS-D2207
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

COMPRESSED AIR SCRUBBER

(D-2207)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-D2207
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 22, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	DEC 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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VESSELS

PROJECT SPECIFICATION

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SHEET 3 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	12APR12	TS	HDT				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.
- F. Suppliers (or equal):

Spray Nozzle:

BETE Fog Nozzle Inc.

www.bete.com

Profile Wire Screens:

Johnson Screens

www.johnsonscreens.com

Nagaoka International Corporation

www.nagaokajapan.co.jp

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15, 3-35, 9-11

UOP Standard Drawing: 3-115, 3-116, 3-122, 3-123, 3-144, 3-183, 3-190, 3-264, 3-266, 3-313, 3-322, 8-121

UOP Project Specification: 963765-106, 963765-307



PROJECT SPECIFICATION

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REV	DATE	BY	APP'D	REV	DATE	BY	APP'D
0	27APR12	TS	HDT				
1	03JUL12	JG	HDT				

REV

VESSELS

SERVICE COMPRESSED AIR SCRUBBER

ITEM NO D-2207

Design	INT 3.5 kg/cm2 (g) @ 170° C
Conditions	EXT ° C
at Top	Metal Temperature(Min) 15° C
Operating	1.5 kg/cm2 (g) @ 44° C
Conditions	° C

Radiograph MR

Postweld Heat Treat MR

Material	Corrosion Allowance
KILLED CARBON STEEL	3

This Vessel is in the Following Service(s):

Hydrogen	-
Hydrofluoric (HF) Acid	-
Hydrofluoric (HF) Acid (with Resid. Elem. Control)	-
Wet Hydrogen Sulfide (H2S)	-
Severe Cyclic	-
Caustic	-
Carbonate	-
Amine	-

NOTES:

Vessel Heads

Top 2:1 ELLIPTICAL

Bottom 2:1 ELLIPTICAL

Accessories Applied by Fabricator

Ladder & Platform Clips	AS REQUIRED
Insulation Clips & Rings	NO
Vessel Support	YES

Nozzles and Manways

Mark	No	Size Inches	Service
AB	1	6	VENTILATION
AI	1	4	INLET (7)
AM	1	24 ID	MANWAY
AO	1	30	OUTLET
AV	1	2	VENT
BI	1	30	INLET
BL	2	2	LG (2)
BM	1	24 ID	MANWAY
CL	2	2	LT (2)
CM	1	14	RELIEF
CO	1	6	OUTLET

Class - ASME CL150
(7) - SPECIAL (2) - ASME CL300

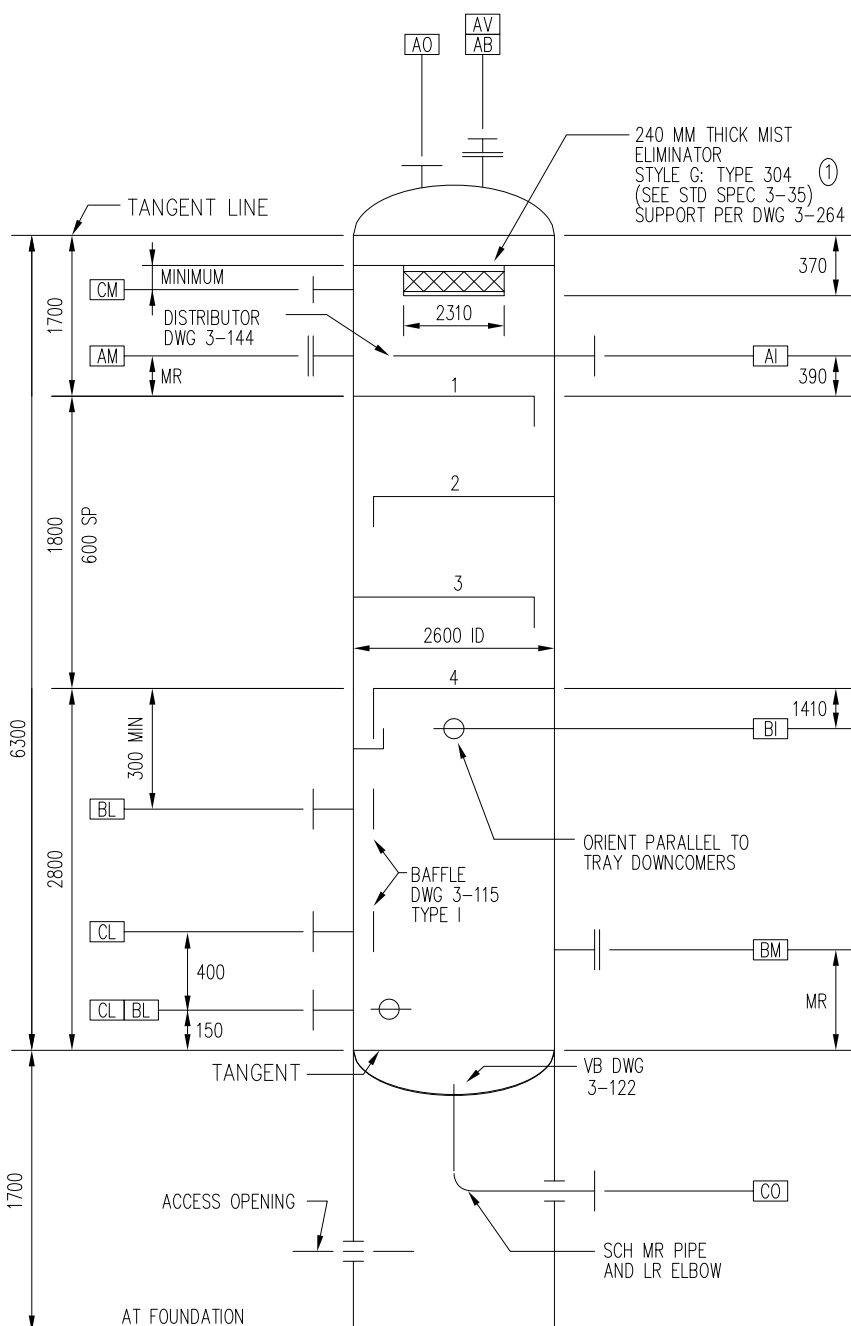
Facing - RAISED FACE

Normal Liquid Level = 350mm ABOVE
BOTTOM TANGENT

Specific Gravity = 0.99

Delta-P Vessel(Total) = 0.04 kg/cm2

NOZZLES MUST NOT BE LOCATED IN DOWNCOMERS



Drawings Referred to in this Specification

3-115	3-122	3-144	3-264			
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Revision
Indication



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**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

PROJECT SPECIFICATION

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SHEET 5 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	05Jul12	HDT	MEH				

**UOP Phenol Process Unit
Oxidation Section**

Item Number	Service	Sheet
	General Notes	6 & 7
D-2207	Compressed Air Scrubber	8
	Tray Data	9

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-18

UOP Standard Drawing: 3-280, 3-311

UOP Project Specification: 963765-301



**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

PROJECT SPECIFICATION

963765 — **307**

SHEET 6 / 9

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	05Jul12	HDT	MEH				

The following general notes apply to all items listed on individual specification sheet:

- A. All items shall comply with the requirements, terms and conditions of this Project Specification and UOP Standard Specification 3-18. For details of vessels see UOP Project Specification 301.
- B. All trays are numbered from **TOP #1 TO BOTTOM**.
- C. Abbreviations: CS = carbon steel; Type 410 = 13% chromium alloy; Type 304 = 18% chromium -8% nickel alloy; Type 316 = 16% chromium – 12% nickel alloy; "M.R." or "MR" = manufacturer's and / or contractor's responsibility. SS may be any of the following – Type 410, Type 304 or Type 316. See UOP Standard Specification 3-18 for additional clarification on materials.
- D. When system of measurement is metric, all dimensions are in millimeters except as noted.
- E. For vessels where the inside diameter is indicated to be minimum (min), adjustments in tray dimensions to accommodate the final inside diameter selected shall be to the chord height of side downcomers which are also indicated as minimum (min).
- F. Vessel nozzle and internals orientation errors are a common source of problems for trayed columns. UOP does not supply specific orientation drawings in order to allow flexibility to the contractor, the vessel fabricator and the internals supplier. The contractor is to ensure sufficient communication between the various parties to avoid orientation difficulties. Some common orientation restrictions follow.
 - Thermowells that are inserted into trayed distillation columns must not contact internals such as support rings or downcomer panels.
 - Reboiler return nozzles should be oriented to minimize vapor interaction with the liquid from the bottom tray traps.
 - Adjacent distillation column trayed sections normally have specific orientation requirements to facilitate vapor and liquid movement between the sections. The supplier of the vessel internals (trays and packing) must provide the vessel fabricator with orientation drawings.
 - Nozzles shall not be located in downcomers.

The following notes apply only when listed for individual item of specification sheet:

1. One pass.
2. Two pass, center downcomers.
- 2x. Two pass, side downcomers.
3. Three pass.
4. Four pass, intermediate downcomers.
- 4x. Four pass, side and center downcomers.
5. Tray(s) shall be liquid tight by seal welding, except at manway(s) where gasketing shall be used.
6. Tray(s) including downcomer(s) shall be liquid tight by seal welding, except at manway(s) where gasketing shall be used.



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**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	05Jul12	HDT	MEH				

7. Do not provide drain holes. Tray(s) shall be liquid tight.
8. Stiffeners and support members transverse to the liquid flow direction shall not exceed 2" (50 mm) in depth and shall not be located above upcomers, or within a lateral distance from upcomers equal to one tray space.
9. Trays shall be assembled on rods and spacers or arranged in cartridges with trap attached to downcomer of bottom tray. Furnish Teflon or metal wall wiper at each tray for liquid seal.
10. All tray assemblies shall be designed for installation and removal from top and bottom.
11. Each tray shall have froth initiator(s) in accordance with UOP Standard Drawing No 3-130.
12. Bubble caps shall be in accordance with UOP Standard Drawing No _____ Type No _____.
13. Ceramic Fiber gasketing shall be used as required for proper design and installation of tray(s), manway(s) and / or distributors for packed beds.
14. Each tray shall have a total of ____ — mm diameter holes on approximately mm equilateral triangular pitch.
15. Each tray shall be designed for an upward pressure differential of ____.
16. Drawoff well(s) on tray No ____ shall be ____ deep and a minimum of ____ wide and extend under entire downcomer area.
17. Provide an additional ____ percent of the required random packing to cover losses.
18. Minimum thickness of tray components is 12 gauge (0.104" / 2.6 mm). A corrosion allowance is not required.
19. Bolts and nuts for trays shall be carbon steel (ASTM A307 Grade A for bolts and A563 Grade A for nuts).
20. Tray No(s) ____ shall be designed for a net uplift of 100 pounds per square foot (488 kilograms per square meter).
Tray No(s) ____ shall be designed for a total downward load of 100 pounds per square foot (488 kilograms per square meter) plus dead weight.
21. Customer/Contractor must verify that the structural and mechanical condition of the existing tray(s) and/or tray supports is satisfactory for continuing operation.
22. UOP MD and Slotted Sieve tray decks, downcomers, installation hardware and distributors; and Raschig Super-Rings and distributors are to be supplied by:

Supplier (Mandatory):

UOP LLC
175 East Park Drive
PO Box 986
Tonawanda, NY 14151-0986
Telephone 716-879-7615
Facsimile 716-879-7215

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**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	05Jul12	HDT	MEH				
1	30Aug12	SA	MEH				

Service: Compressed Air Scrubber

Item Number: D-2207

Column Assemblies

Item No	Quan Assem	ID of Vessel	Material	Description	Dwg or Sht No	Dimensions			Tray No	Notes (Sheet 6&7)
1	4	2600	CS	Valve Trays	7				1 – 4	1
2	1	2600	CS	Trap	3-280	N = MR	S = *		Below Tray 4	

* See sheet 9

Tray sections shall pass through a 24" ID vessel manway.

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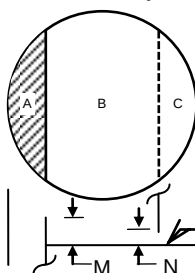


**TRAYS AND PACKING
(RANDOM AND STRUCTURED)**

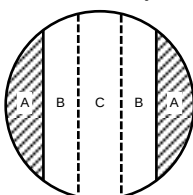
REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	05Jul12	HDT	MEH				
1	30Aug12	SA	MEH				

Column: Compressed Air Scrubber		Item Number: D-2207		
Tray Data				
Tray Identification (Tray Numbers)	All (1 - 4)			
Vessel ID, mm	2600			
Operating Pressure, kg/cm ² (g)	1.5			
Operating Temperature, °C	42			
Structural Design Temperature, °C				
Fluid Description	Air – 0.1% Caustic			
Foam Derating Factor	1.000			
Vapor -Flow Rate, kg/h	61,539			
Density, kg/m ³	2.8			
Liquid -Flow Rate, kg/h	28,208			
Density, kg/m ³	992			
Viscosity, cP	0.63			
Surface Tension, dynes/cm	69.2			
Type of Tray	Valve			
For Sieve Trays				
Weir Height, M, mm				
Downcomer Clearance, N, mm				
Total Number of Holes				
Hole Diameter, mm				
Hole Area/Bubbling Area x 100				
Tray spacing, S, mm (for exceptions see Project Specification 301 Vessels)	600			
Notes				
Tray Layout				
Number of Passes	1			
A, mm	208			
B, mm	2184			
C, mm				
D, mm				
E, mm				
F, mm				
G, mm				

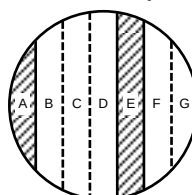
1-Pass Tray



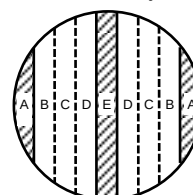
2-Pass Tray



3-Pass Tray



4-Pass Tray



Maximum allowable overall pressure drop for vessel is 0.035 kg/cm²

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